ROUTE	GARAGE	BODY	PERIOD
70 Grand	Brentwood	Task force baseline	Jul 2023 – Jul 2026

COMPILED	STATUS
Aug 19, 2026	DRAFT

RUN GUIDE: WHERE ROUTE 70'S MINUTES GO

Route 70 Grand is Metro's busiest bus line: about 135,000 boardings a month. The COO's charter asks the task force to move on-time performance from roughly 77% at charter signing (July measured: 75.3%) to at least 85%. This run guide shows where the late minutes are actually born, using the same data every department already trusts.

75.3%

OF CHECKPOINTS ON TIME, JULY 2026 — GRADING ONLY THE SERVICE THAT OPERATED.
COUNTING MISSED TRIPS AGAINST THE SCORE: **72.9%**

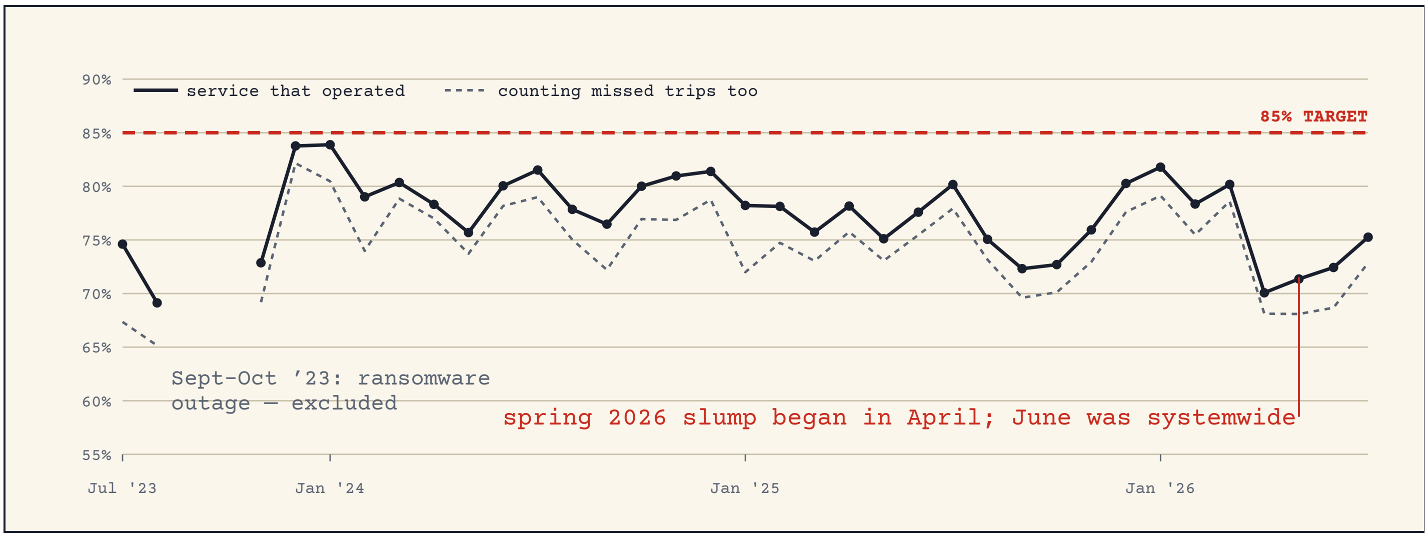
85% REQUIRED

→ *the gap is 10–12 points. Most of it is decided before the bus moves.*

THREE YEARS OF THE SAME STORY

Each dot is one month of Route 70. "On time" means the bus crossed its scheduled checkpoints no more than 5 minutes late and no more than 1 minute early, applied at every checkpoint, measured trips only (never estimated). Two honest ways to count: the solid line grades only the service that operated; the dashed line also counts every missed trip against the score. The space between them is the missed-service penalty — when it widens, buses are not just late, they are not coming.

drag sideways for the full chart →

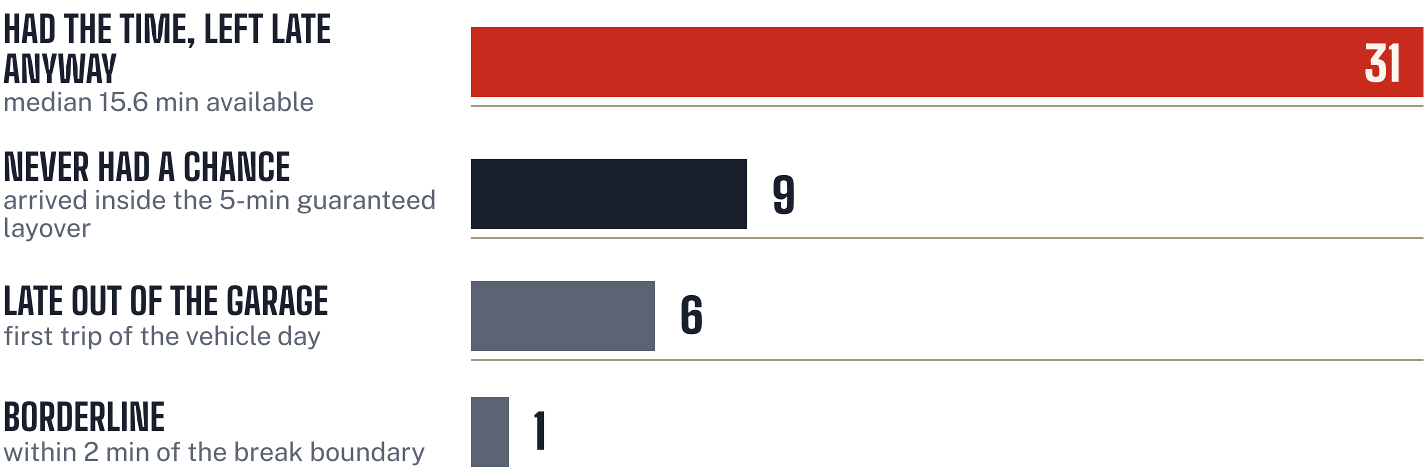


Monthly on-time performance, Route 70, from the checkpoint-level AVL record (39 months, ~155,000 trips). Solid: on-time crossings ÷ crossings served (operated service only). Dashed: on-time crossings ÷ crossings scheduled minus waived (missed service counts against the score). September-October 2023 are excluded: a ransomware event took the AVL record down (3 days and ~40% of days captured, systemwide). The spring 2026 slump began in April (70.1% operated-service, the worst full month in the record) and ran through June; June's dip is known to be systemwide, and whether April-May were is not yet checked.

ONE DAY UNDER THE MICROSCOPE

Tuesday, July 8, 2026, the day the service planners worked through trip by trip. Route 70 ran 168 trips. 47 of them left their first checkpoint more than two minutes late, the departure rule the task force adopted on August 19. The question that decides everything: **did the bus have time to leave on time?** Every trip's terminal arrival, layover, and departure is in the AVL record, so we can answer it for each one.

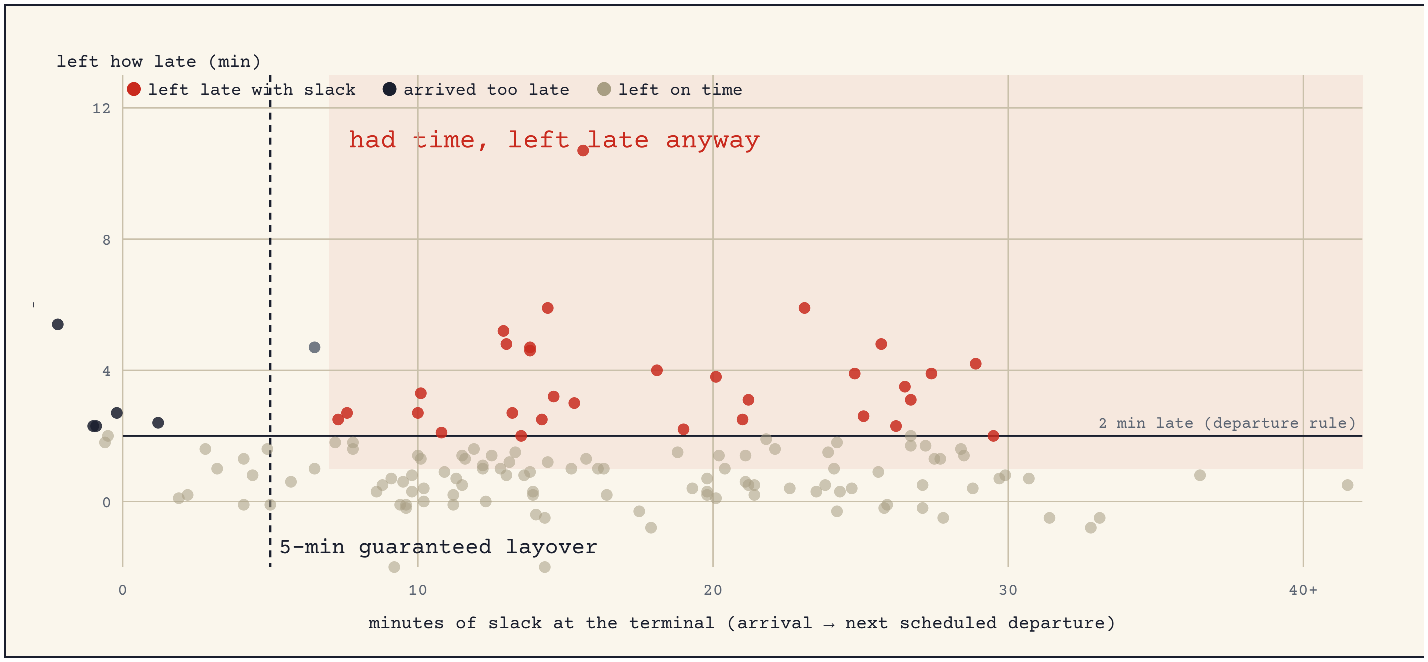
47 of 168 trips left their first checkpoint more than two minutes late on July 8. Where they came from:



Had the time, left late anyway (31 trips). These buses reached the terminal with a median of 15.6 minutes before their next scheduled departure — three times the contractual 5-minute break — and still left more than two minutes late, taking a median 21.2 minutes of terminal time. This is the piece supervision and dispatch can move without touching the schedule.
Never had a chance (9 trips). These arrived with less than 5 minutes remaining — the contract's guaranteed-layover minimum — so the lateness was inherited from the previous trip. More discipline cannot fix these; only more recovery time or faster running can. (The classification grants the 5-minute allowance at every terminal, which is more generous than the contract's guarantee of 5 at one end per round trip — so the "had the time" counts are, if anything, understated.)
Late out of the garage (6 trips). First trip of the vehicle's day, already late at the first checkpoint. The radio log confirms it: three "out late — no bus" calls in the 5 p.m. hour alone, one 35 minutes late.

→ same symptom, opposite remedies. Discipline where slack was wasted; schedule where slack never existed.

drag sideways for the full chart →



Every July 8 trip with a measured terminal turn. Horizontal: minutes between arriving at the terminal and the next scheduled departure ("slack"). Vertical: how late the bus actually left. The shaded region is the uncomfortable one: plenty of slack, late anyway. The contract guarantees at least 5 minutes of layover at one end and 10 minutes per round trip (Section 8); scheduled layover beyond the guarantee is recovery meant to absorb lateness.

A YEAR SAYS THE SAME THING

The one-day finding could have been a fluke of the day Planning picked. It isn't. A checkpoint-level pull of the last twelve months — 55,249 measured Route 70 trips across 383 service days — repeats the July 8 story almost exactly, month after month.

37%

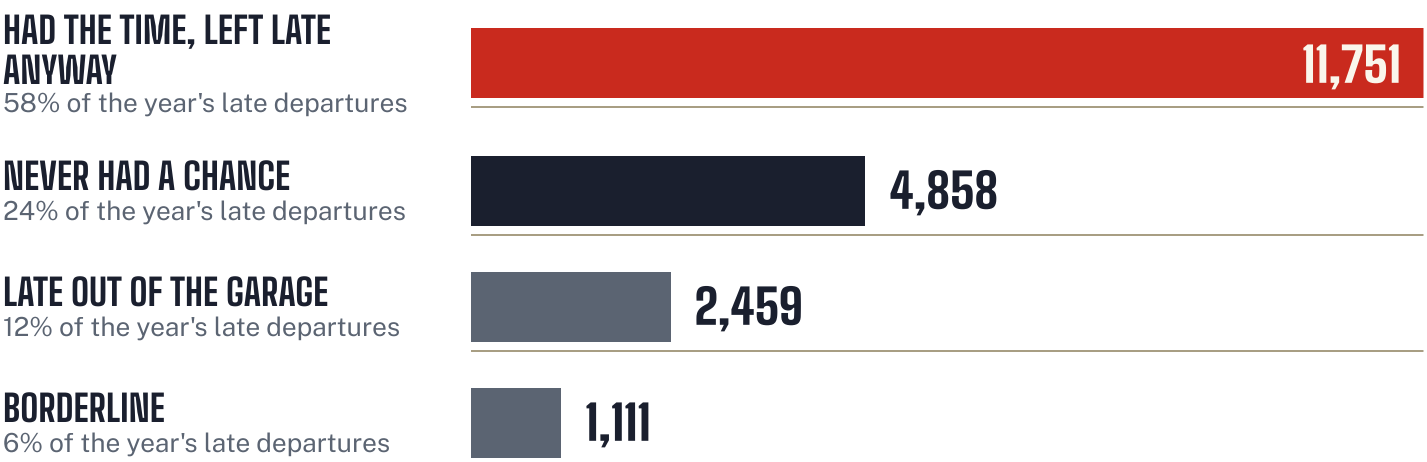
OF TRIPS LEFT THE FIRST CHECKPOINT >2 MIN LATE (YEAR)

58%

OF THOSE HAD 7+ MINUTES OF SLACK AND LEFT LATE ANYWAY

18

MEDIAN MINUTES OF SLACK WHEN IT HAPPENED



The two ends of the route tell opposite stories. When a bus leaves North Broadway Transit Center late, slack was sitting there 79% of the time — a supervision and dispatch question. When a bus leaves Loughborough Commons late, slack was available only 37% of the time — much more of the south end's lateness is inherited from tight turns, a schedule question.

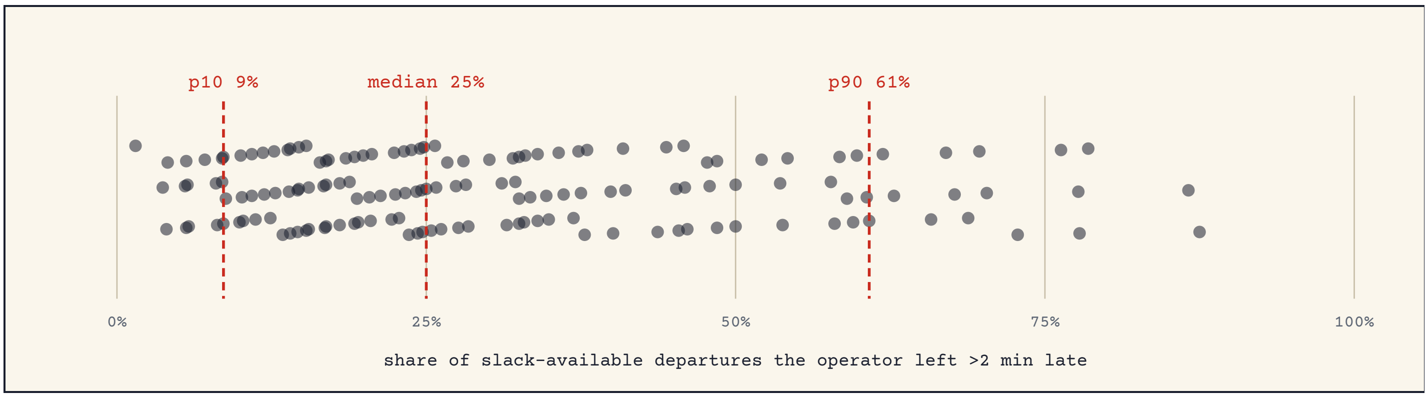
LOUGHBOROUGH COMMONS
9,813 late departures

→ one route, two remedies: sort charging holds from behavior at the north terminal, re-time the south turns.

SAME RUN, DIFFERENT OPERATOR

Is this about individual operators, or about the runs they're assigned? The twelve-month data can separate the two, because most runs are driven by different operators on different days. Looking only at departures where leaving on time was clearly possible (7+ minutes of slack): the spread between operators is wide, and it persists on the *same run*. No names or badge numbers appear here or anywhere in this document; identifiers are anonymized.

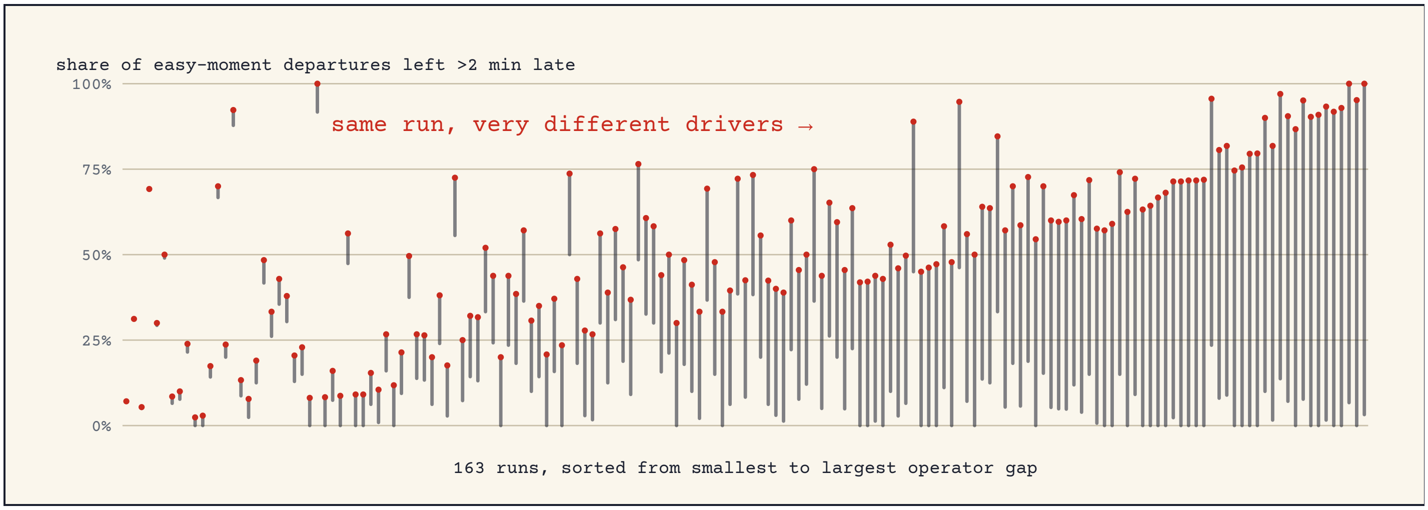
drag sideways for the full chart →



Each dot is one operator (153 with at least 30 slack-available Route 70 trips in the year). Position: how often that operator left more than two minutes late when 7+ minutes of slack were available. Rates are moderated toward the group average for operators with fewer trips, so a thin sample cannot produce an extreme dot. The same-run test: 163 runs (the scheduled daily assignments, like the run numbers in the radio log below) were driven by two or more different operators this year. Comparing operators on the identical assignment, the median gap between the most and least late-leaving operator on a run is 34 points; assignment does not explain the spread.

Read it as two drivers sharing one assignment: the same run, driven by different operators on different days, is the same schedule, the same terminals, the same traffic. If the run were the cause, the two drivers would look alike. They don't — and the chart below draws that directly. Each vertical line is one run; its ends are that run's most and least punctual operators (each with at least 10 easy-moment departures). The line's length is the disagreement between people driving the identical assignment.

drag sideways for the full chart →



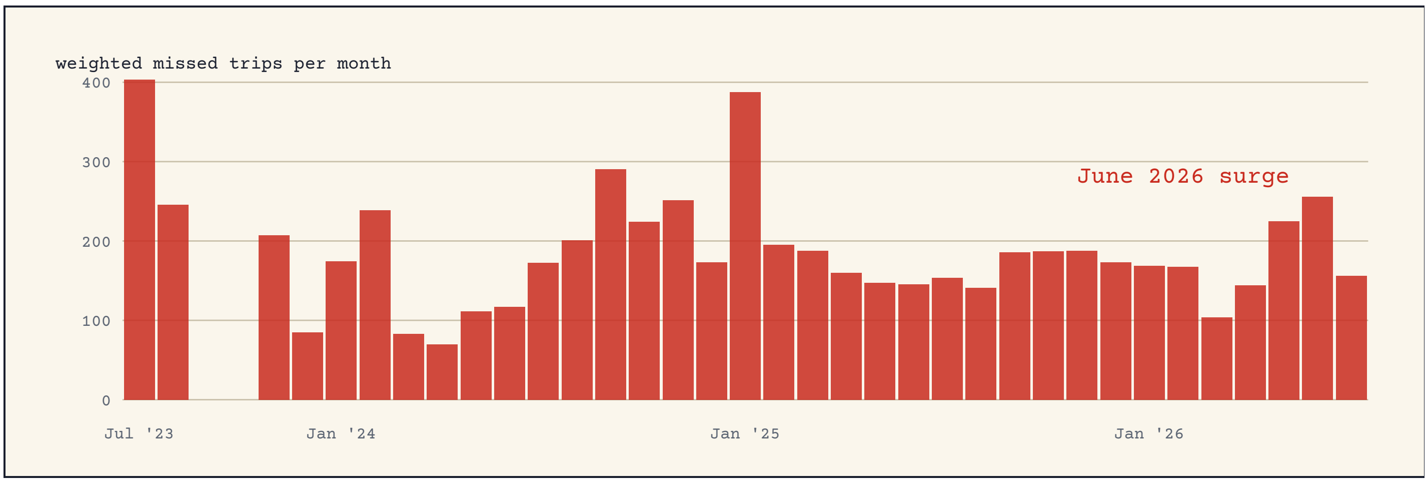
Every run driven by two or more operators this year (163 runs, sorted by gap). Bottom of each line: that run's most punctual operator's late-with-slack rate. Top: its least punctual operator's. Long lines everywhere mean the assignment does not explain the behavior.

Could the radio explain it? We checked. "Left late with slack available" is measured opportunity, not proven intent — so every one of the year's 11,751 slack-available late departures was matched against the Trouble Log (same operator, same service day, incident logged in that terminal window). Only **5.9%** have any logged incident at all, and only **1.3%** have an external one — traffic, mechanical, or passenger-related. The remaining **94.1% are radio-silent**. Delays under 10 minutes are not logged, so small events are invisible — and one known cause leaves no radio trace at all: at North Broadway, a bus still on its charger can hold a departure without any incident being logged, so part of the north end's pool may be charging, not conduct. Charger session data would separate the two; the south end and the same-run comparison below carry no such confound. This is the pool coaching and terminal supervision can move, and at the observed spread, moving the top quarter of the distribution toward the median is worth several points of route OTP on its own.

THE SERVICE THAT NEVER CAME

On-time performance only grades buses that showed up. Weighted missed service counts the ones that didn't: each trip scored by how much of it was lost: a trip that never ran counts 1, a trip missing most of its checkpoints counts 0.75 or 0.5, a trip missing one counts 0.25, a complete trip counts 0. It is the house's primary missed-service measure; the plain count of fully-missed trips is secondary. To be clear about proportion: missed service is *not* Route 70's dominant failure — lateness is — and it costs about 4% of a typical weekday's service. But it is concentrated where it hurts most, and its main cause (buses leaving the garage late or not at all) is among the most directly fixable findings on this page.

drag sideways for the full chart →



Weighted missed trips per month, Route 70 (same 39 months as the trend above). This is the quantity that separates the two lines in the first chart.

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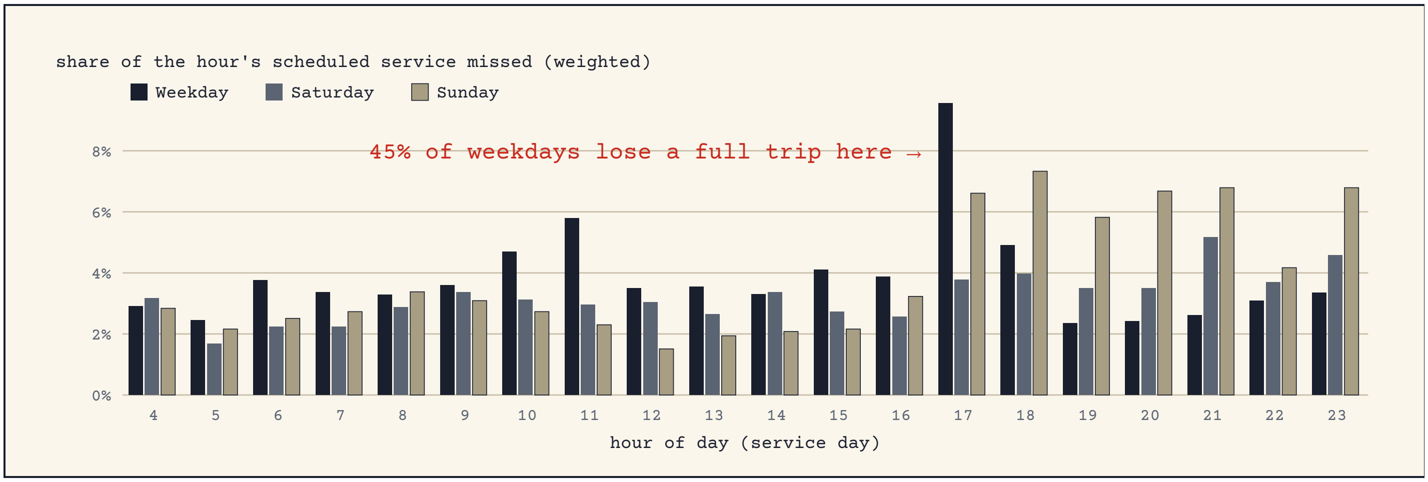

Each bar: the weighted missed trips in that hour as a share of the trips scheduled in it, by schedule day type (a holiday counts as the schedule it ran), 2025-08 through 2026-07 (months with extracts; 52 Saturday days, 58 Sunday days, 252 Weekday days). A share reads the same for a 9-trip peak hour and a 4-trip evening hour, so the three day types and all hours compare on one scale. The absolute mass is the monthly chart above. The tallest bar is where riders lose the most service — not necessarily the hour with the worst on-time performance.

SAME DAY, THREE ANSWERS

The service planners' investigation tool and the R&D pipeline disagree about Route 70's numbers, but it's a definitions difference, not a data conflict. The tool counts a trip late if it leaves its first checkpoint more than **1 minute** late (measured from its own output; the margin notes argue for 2). The standard band doesn't count a departure late until **5 minutes**. Pick the rule, and July 8 gets a different verdict. The task force settled it on August 19: **departures are judged at 2 minutes; overall OTP keeps the production 5-minute band**:

Share of the 168 trips leaving the first checkpoint late, by rule: >1 min → 55% · >2 min (adopted) → 28% · >5 min → 7%.

Before the task force reads any two reports side by side, it needs one stated rule for the first checkpoint. Until then, a 55%-late day and a 7%-late day can be the same Tuesday.

WHAT THE STREET CONTRIBUTED

If the schedule's driving times were too tight, buses would lose minutes between checkpoints. On July 8 they mostly didn't: on nearly every segment the typical bus ran *at or faster than* schedule. The street shows up in the afternoon, and in specific places.

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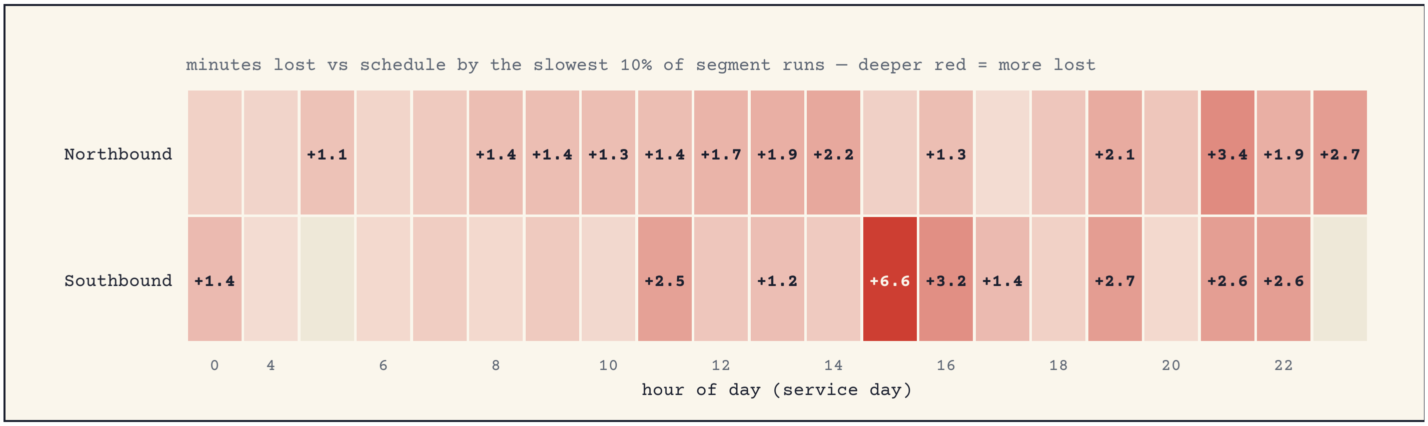

By hour and direction: minutes lost versus schedule by the slowest tenth of segment runs (a surge measure; the typical run is shown in the ledger above, because a median would hide the afternoon). Radio log confirms the hotspots: construction closure at Grand & North Market (11:30), traffic onto I-44 (afternoon), Grand & Chippewa, Grand & Park.

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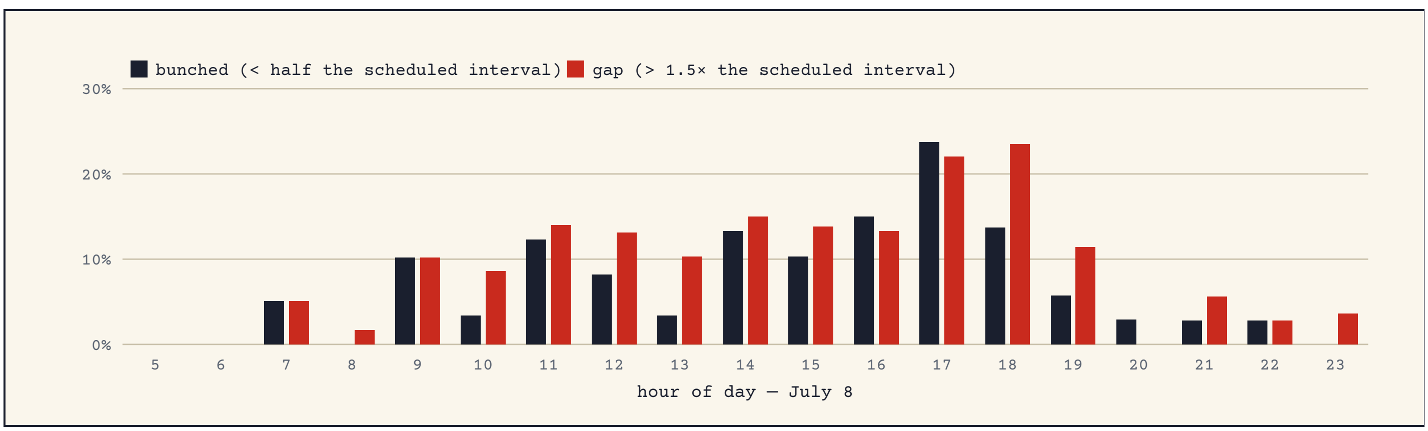
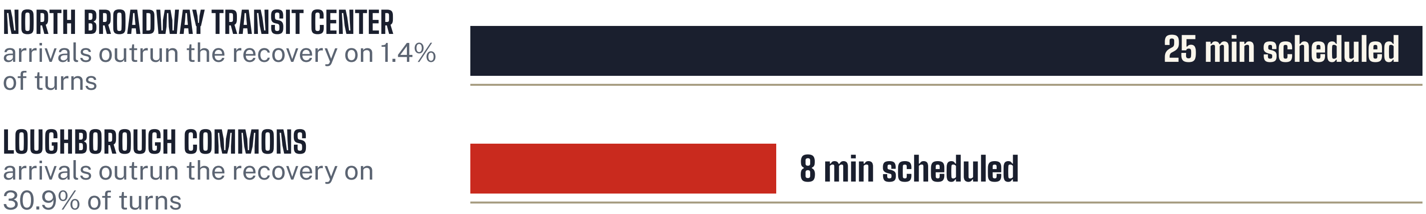


What riders felt: of 978 measured intervals between consecutive Route 70 buses on July 8, 7.6% were bunched (bus arrived less than half the scheduled interval behind the one ahead) and 9.7% were gaps (more than 1.5× the scheduled interval — for a 15-minute schedule, a wait over 22 minutes).

IS THE SCHEDULED RECOVERY ENOUGH?

The scheduling system answers the other half of the terminal question. Each round trip carries layover — schedule slack at the ends meant to absorb lateness — with a contractual floor (current ATU agreement, Section 8: at least 10 minutes of layover per round trip, with 5 minutes guaranteed at one end; pull-out and turn-in trips excepted). Pulling the current weekday schedule from the scheduling system and setting it against a year of measured arrivals:

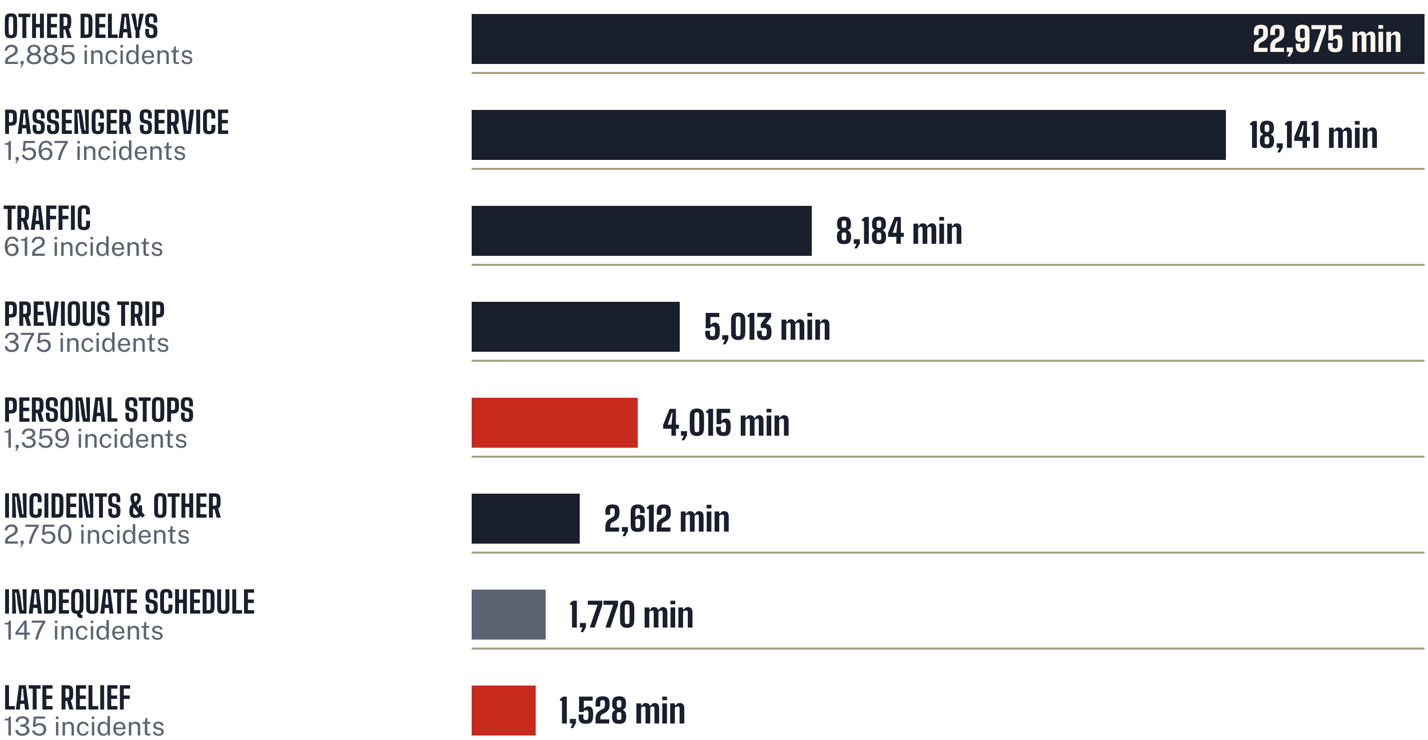


The schedule banks nearly all of its layover at the north end — and much of that is charging time, not idle slack. The electric-bus chargers are at North Broadway, so its median 25 minutes of scheduled layover serves the buses as much as the operators; it cannot simply be shifted south. Only **1.4%** of arrivals there outrun what the schedule can absorb. Loughborough turns get a median 8 minutes, and **31%** of arrivals eat past what the schedule can absorb — the south end's inherited lateness is a schedule-design outcome, not an operator one. (The split is contract-compliant, and Section 8 permits any split meeting the 10-minute round-trip and 5-at-one-end floors — but for battery-electric blocks the charging requirement pins the layover north, so relieving the south end likely means adding time there rather than moving it.)

→ the terminal split, explained: the north end's layover is pinned by the chargers; the south end's thin recovery is overwhelmed by arriving lateness.

WHAT THE RADIO HEARD

Over the last twelve months the Trouble Log recorded **10,507** Route 70 incidents carrying **65,781** logged delay minutes. Delays under 10 minutes aren't logged, so every figure here is the floor, not the total. Where the logged minutes went:



And one day in full: every Route 70 trouble call logged on July 8, the day under the microscope above:

TIME	RUN	PROBLEM	WHERE	DELAY
03:24	2521	**Out Late-Miscellaneous**	Brentwood garage	**10 min**
06:07	2530	**Out Late-Miscellaneous**	Brentwood garage	**11 min**
11:30	2532	Blocked Const/Road Closure	at N GRAND BLVD and N MARKET ST	—
14:52	2927	Inadequate Schedule	at N GRAND BLVD and LINDELL BLVD	**16 min**
15:48	2532	Traffic	at E GRAND BLVD and N 19TH ST	**13 min**

TIME	RUN	PROBLEM	WHERE	DELAY
15:51	2541	Blocked-Accident/Emerg Equip	Last Time Point: B.	—
16:47	2532	Previous Trip	at S GRAND BLVD and UNNAMED ROAD	**12 min**
16:57	2535	Traffic	at S GRAND BLVD and CHIPPEWA ST	—
17:02	2541	Traffic	83 yards SSE of LOUGHBOROUGH DR and S GRAND DR	**16 min**
17:05	2537	Passenger Service	59 yards N of S GRAND BLVD and HICKORY ST	**10 min**
17:05	2541	Previous Trip	at LOUGHBOROUGH COMMONS DRIVEWAY and LOUGHBOROUGH COMMONS SHOPPING CENTER PRIVATE ROA	**12 min**
17:08	2544	**Out Late-No Bus**	BRENTWOOD METROBUS FACILITY	**35 min**
17:17	2543	**Out Late-No Bus**	Brentwood garage	**11 min**
17:18	2533	Traffic	at S GRAND BLVD and PARK AVE	**10 min**
17:28	2534	Traffic	at S GRAND BLVD and SCOTT AVE	**10 min**
17:31	2545	**Out Late-No Bus**	Brentwood garage	**10 min**
18:03	2543	Previous Trip	GRAND ARSENAL	**12 min**
18:29	2538	Passenger Service	at GRAND METROLINK STATION WALKWAY and SCOTT AVE	**11 min**
18:44	2541	Passenger Service	61 yards N of S GRAND BLVD and CHOUTEAU AVE	**11 min**
21:28	2543	Passenger Service	at S GRAND BLVD and CAROLINE ST	**10 min**
22:49	2541	Passenger Service	at S GRAND BLVD and HICKORY ST	**10 min**
23:10	2543	Personal Stop	at E TAYLOR AVE and N BROADWAY	**10 min**

HOW TO READ THESE NUMBERS

ON TIME (STANDARD BAND)	A checkpoint crossing no more than 5 minutes late and no more than 1 minute early (−300 to +60 seconds), waived checkpoints excluded. Measured trips only, never estimated.
TWO OTP MEASURES	**Operated-service OTP** (the house "traditional" measure) = on-time crossings ÷ crossings served — it grades only buses that ran. **Missed-service-counted OTP** (the house "true" measure) = on-time crossings ÷ (crossings scheduled − waived) — a trip that never ran counts against it. They are linked by an identity: true = traditional × share of scheduled service delivered, so the gap between the two lines is exactly the missed-service penalty. July 2026: 75.3% vs 72.9%. Which measure the charter's 85% target is scored on is a decision the task force should make explicitly.
DEPARTURE RULE (ADOPTED AUG 19, 2026)	Leaving a layover point or the first checkpoint is judged late past **2 minutes**: the task force's decision, matching what the service planners' margin notes proposed (their tool had been using 1 minute; the standard band waits until 5). Every departure-ledger figure on this page uses the 2-minute rule; overall OTP keeps the production 5-minute band. The rule picker above shows how the verdict moves across all three thresholds.
SCOPE OF EACH FIGURE	Trend: 39 months, ~155,000 Route 70 trips (Sept–Oct 2023 excluded for the ransomware outage). July 8 ledger, street, radio: one weekday, chosen by the service planners — 168 trips, 996 checkpoint crossings. Year-scale ledger and operator sections: Aug 2025 – Aug 18 2026, 345,000 checkpoint crossings, 55,249 measured trips, 383 service days, operator badge resolved on 99.3% of crossings.
LAYOVER (THE CONTRACT)	ATU Local 788 O&M Agreement 2022–2025 (board-approved), Section 8.A: schedules provide layover at both ends to the maximum extent practicable, with a layover required at one end; **a minimum 5 minutes is guaranteed at one end of each round trip**, and **total scheduled layover per MetroBus round trip is at least 10 minutes** (pull-out and turn-in trips excepted; round trips of 40 minutes or less are exempt from the 10 but must give 5 at one terminal end within 40 minutes of operation). The 2025 successor agreement is in negotiation; the current update materials do not touch layovers.
SOURCES	AVL timepoint record (TMDataMart ADHERENCE), including the Aug 2025–Aug 2026 checkpoint pull of 2026-08-19; Trouble Log incident tables, same window; Trapeze FX weekday schedule, June 15 2026

pick (scheduled recovery per trip); tidy trip extracts Jul 2023–Jul 2026; OTP Investigation Tool v1.3 export, Jul 8 2026 (L. Peterson); MetroBus Trouble Record, Jul 8 2026; Route 70 Improvement Task Force charter (R. Forrest, Jul 29 2026).